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Land Use/Land Cover changes dynamics and their effects on Surface Urban Heat Island in Bucharest, Romania

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ABSTRACT

The aim of the paper is to examine and analyze the long-term spatial dynamics of LULC (Land Use/Land Cover) and the impact of their changes on the SUHI (Surface Urban Heat Island). The study covered the area of Bucharest Municipality, the capital of Romania and its neighboring areas, and involved a series of Landsat images, taken in the summer, and ranging from 1984 to 2016. To achieve the goal the following were accomplished: (i) a supervised image classification, done with good accuracy and (ii) a post classification change detection approach was used to assess the LULC changes; (iii) the thermal signature for each land use category was determined using LST (Land Surface Temperature); also, iv) NDVI (Normalized Difference Vegetation Index) has been used to determine a relationship between the thermal behavior and the characteristics of each land use category. The results have revealed that by 2016 there has been an increase in the built-up areas and the fallow land areas and a decrease in the croplands and forested areas. Also, there were observed the increase in surface temperature and a decrease for the difference between the average temperature of the urban area and the surrounding areas of the city indicating the extension of the area where the phenomenon of the Urban Heat Island of Bucharest occurs.

1. Introduction

Along with the evolution of human society, cities have emerged and expanded as a natural consequence. According to United Nations statistics, more than half of the world/global population lives in urban areas and forecasts indicate an increase in this percentage in the future (United Nation, 2015).

The phenomenon of urbanization, viewed both from the point of view of people's migration to cities and from the point of view of the transformation of the land cover, was closely accompanied by the environmental pollution phenomenon, concerning in particular on waste (industrial and household) and air pollution and, finally, that of climate change.

By modifying the characteristics of the land surface (e.g., albedo, emissivity and roughness etc.), cities influence the regional heat flow (Li et al., 2014), the local flow (Bowen, 1985; Hanna et al., 2007), circulation (regional heat flux, local airflow and circulation) (Collier, 2006), turbulence in the boundary layer (Ludwig and Dabberdt, 1973), rainfall (Chen et al., 2015) and the thermal regime (Cai et al., 2017).

The urban areas can have higher temperatures than the surrounding suburban/rural areas, this phenomenon is known as Urban Heat Island (UHI) (Berkowicz and Prahm, 1984) and it was first observed by Luke Howard (1772–1864). Since then until today, a very large number of articles (Baklanov et al., 2016; El-Hattab et al., 2018) and reviews (Deilami et al., 2018; Mohajerani et al., 2017; Ward et al., 2016; Yan et al., 2016) related to the Urban Heat Island were published.

Numerous research studies on the heat island effect have demonstrated the existence of several types of connections between the UHI and various environmental variables (Chapman et al., 2017; Deilami et al., 2018; Zhang and Sun, 2019). These were very well summed up in Arnfield's work (Arnfield, 2003), from which we exemplify: UHI – decreases with the wind speed increasing, – decreases with the nebulosity increasing, – is more intense in anticyclonic regime, – is more intense at night, – the intensity increases with the increase in size of the urban area and population. Recent concerns have also been related to the relationship between land use land cover change and land surface temperature (LST) (Kubota et al., 2017; Tran et al., 2017; Yu et al., 2018).

This paper is intended to be a completion to the research work on the urban heat island of Bucharest, the capital of Romania. Further investigations were done both in studies across several European cities (Pongrácz et al., 2010; Schwarz et al., 2011; Ward et al., 2016) as well

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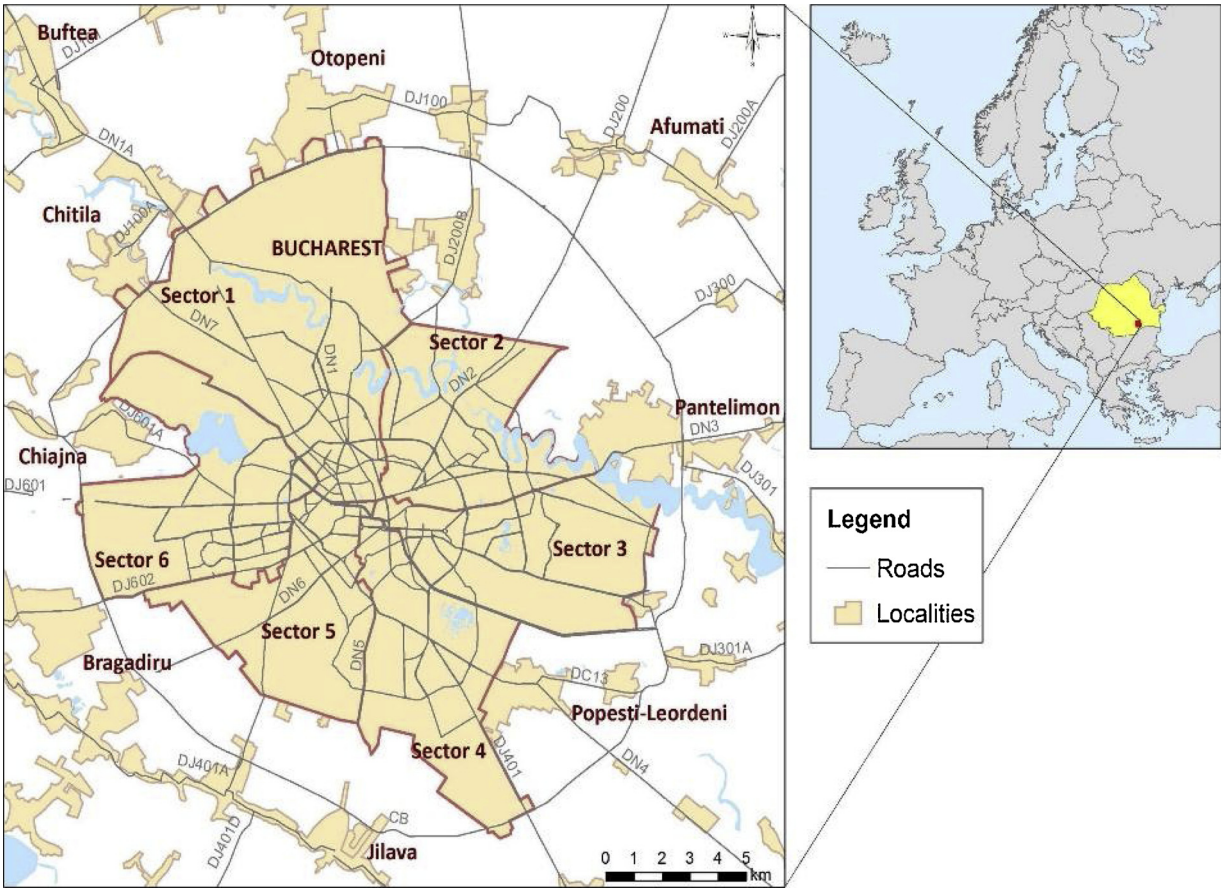


Fig. 1. Location of analyzed area.

as in studies covering the area surrounding Bucharest. The latter were based on *in situ* temperature measurements at fixed and mobile locations (Tumanov et al., 1999), or temperatures obtained from MODIS satellite products (Moderate Resolution Imaging Spectroradiometer) (Cheval and Dumitrescu, 2009) and have highlighted the existence of the Bucharest Heat Island, its expansion both during the day and during the night, and the connection to the type of land cover (Cheval and Dumitrescu, 2015), the daily and sub-daily thermal variations (Cheval and Dumitrescu, 2017).

The current study presents a comprehensive analysis of the spatio-temporal evolution of some types of surfaces in the urban area of Bucharest and the influence of these changes on urban microclimate (considering as a characteristic – the UHI) using Landsat satellite data from different years, which have a higher spatial resolution compared to MODIS (previously used data in UHI-Bucharest study).

2. Data and methods

2.1. Study area

The study area, represented in figure Fig. 1, includes the city of Bucharest and the surrounding areas, covers 625 km^2 and it is located between 44°19′12″N to 44°33′36″N and 25°56′24″E to 26°15′36″E.

Bucharest is the capital of Romania, it has a total area of approximately 238 km^2 and 2.1 million residents. The city is located in the southeast part of Romania, it has continental climate, with four seasons (spring – March, April, May, summer – June, July, August, autumn – September, October, November, winter – December, January, February). July and August are the warmest months of the year, with values of 30 °C for the average maximum temperatures. August has the lowest number of rainy days in the year and the smallest amount of rainfall in the summer.

2.2. Used data

Landsat data were considered appropriate for this study because: the program provides the longest series of continuous surface records; data are free and have sufficient spatial resolution (30 m) to highlight the process and the effects of urbanization. In this study there were used satellite images from: Landsat 5 TM (Thematic Mapper), Landsat 7 ETM + (Enhanced Thematic Mapper Plus) and Landsat 8 OLI (Operational Land Imager) since 1984 to 2016, downloaded from the Global Visualization Viewer that belongs to U.S. Geological Survey (USGS, 2018a). To make a better comparison, there were searched images from the summer season in the same period of the year, under clear sky conditions. As an image covering the analyzed area is generated every 16 days and most are cloud-covered, according to the above criteria, 5 images were selected (Table 1).

Table 1
Landsat acquisition date, sensor type, resolutions of products and cloud cover.

Date and time (UTC) of acquisition	Landsat spacecraft and scanning sensor	Resolution (m)	Cloud cover
04.08.1984 – 08:32	Landsat-5 TM	30	0%
01.08.1991 – 08:20	Landsat-5 TM	30	0%
04.08.2001 – 08:46	Landsat-7 ETM +	30	0%
02.08.2009 – 08:46	Landsat-5 TM	30	0%
05.08.2016 – 08:56	Landsat-8 OLI	30	0%

2.3. Methodology

In this study, radiometric and atmospheric calibrations were conducted with the image-based dark-object subtraction method (Chander et al., 2009; Li et al., 2012; Lu et al., 2015). To analyze the Land Use/Land Cover Changes Dynamics and their Effects on Urban Heat Island, two series of operations were carried out:

- *LCLU changes detection and analysis* in which the selected satellite images were analyzed by performing the following operations: (i) a supervised land-cover classification has been achieved (ii) an accuracy assessment was then performed; (iii) the LCLU changes were detected.
- *LCLU changes impact analysis*, required the following operations: (iv) LST determination and study of the relationship between LST and classified images (v) determination of NDVI and analysis of the linear correlation between NDVI and LST.

2.3.1. Land-cover supervised classification and land cover change

There are free data for land use/land cover classifications, like Corine Land Cover and The Urban Atlas, provided by the European Environment Agency, but because there are differences between the two data sets (Petrişor and Petrişor, 2015), the authors of this article have made their own classification of the studied area so that it may better serves the intended purpose.

Satellite data has been pre-processed for georeferencing, mosaic, and extraction of the interest area using ArcGIS Software.

Based on its spectral signature, each pixel in the image has been assigned to the class that fits best. The six defined classes were: water bodies (water) = lakes, rivers; forest = high dense vegetation (trees); built-up areas (built-up) = residential, commercial, industrial, public areas; croplands = cultivated agricultural land; fallow-lands = uncultivated agricultural land; other types (other) = urban green areas, construction sites, inland wetlands, bare land.

After applying the maximum likelihood algorithm, a post-classification controlled primarily by an analyst was applied, in which the analyst selected the representative pixels for the desired classes. The problem of the Landsat mixed pixels (heterogeneous mix of features that mainly include buildings, grass, roads, soil, trees, water) has been approached by visual interpretation, thus increasing the accuracy of its classification and improved the results obtained using the supervised classification.

Land cover/land use classification was made using Landsat satellite images, bands 4-3-2 (Color Infrared) and 3-2-1 (True Color) for Landsat 5 and 7, respectively bands 5-4-3 (Color Infrared) and 4-3-2 (True Color) for Landsat 8.

The post-classification comparison method (Singh, 1989) has been used to detect changes in land use, obtaining maps which presents a complete array of changes from a period of time (Fig. 4).

An evolution over time has been made for land use change by calculating the total surface covered by each category of land in the area under consideration for each reference year.

2.3.2. Accuracy assessment

Using a confusion matrix (error matrix), an accuracy assessment for supervised technique was carried out to determine the degree of correctness of the classification.

The accuracy assessment was performed comparing the categorized pixels from Landsat images with Military Topographic Maps (1:25.000) from the Geoportal of Military Topography Division and with pixels from high-resolution imagery available through Google Earth (test pixels or the ground truth). For each analyzed class 50 test pixels were taken resulting a total of 250 test pixels. The accuracy parameters determined from the matrix of confusion are: user's accuracy, producer's accuracy (Olofsson et al., 2014), overall accuracy and the kappa coefficient (Bogoliubova and Tymków, 2014).

The user's accuracy (*Accuracy*) is calculated by dividing the number of correctly classified pixels in each class by the total number of pixels that were classified in that class, (indicates the percentage of correctly classified pixels, out of the total those identified as belonging to the same class).

The producer's accuracy (*Reliability*) is computed for each class as the number of correctly classified pixels divided by the total number of test pixels of that class (indicates the percentage of pixels in the image that belong to a particular class) (Olofsson et al., 2014).

The *overall accuracy* is calculated as the total number of correctly classified pixels (diagonal elements in confusion matrix) divided by the total number of test pixels (the ground truth).

The Kappa's coefficient (K) is used to determine the agreement between the classified images and reality with values between 0 and 1, where 0 represent complete randomness and 1 perfect agreement. Kappa coefficient was calculated according to equation 1 (Bogoliubova and Tymków, 2014):

$$\hat{K} = \frac{N\sum_{i=1}^{m} D_{ij} - \sum_{j=1}^{m} R_i C_j}{N^2 - \sum_{j=1}^{m} R_i C_j} \quad (1)$$

where K – Kappa coefficient, N – total number of pixels, m – number of classes, ΣD_{ij} – total diagonal elements of an error matrix (the diagonal elements in the matrix represent the number of correctly classified pixels of each class), R_i – total number of pixels in row i (the rows correspond to classes in the ground truth map (or test pixels set)), C_j – total number of pixels in column j (the columns correspond to classes in the classification result).

2.3.3. Retrieval of Land Surface Temperature (LST)

Land surface temperature was obtained using satellite products resampled at 30 m from: band 6 (Thermal Infrared) for Landsat 5 and Landsat 7; band 10 (Thermal Infrared 1) for Landsat 8.

The Landsat images contain Digital Numbers (DN), therefore to obtain surface temperature values, the data will be processed by following a series of steps (Choudhury et al., 2018). The first step is to calculate TOA Spectral Radiance from digital numbers. For Landsat 5 (TM) and Landsat7 (ETM+) the process is differently compared with Landsat 8 (OLI) as follows:

(i) for Landsat 5 and Landsat 7 (USGS, 2018b), Eq. (2):

$$L_\lambda = ((LMAX_\lambda - LMIN_\lambda)/(QCALMAX - QCALMIN)) \times (QCAL - QCALMIN) + LMIN_\lambda \quad (2)$$

where:

(ii) L_λ – TOA Spectral Radiance (W/(m^2 × sr × μm));

(iii) $LMAX_\lambda$ – the spectral radiance that is scaled to QCALMAX (W/(m^2 × sr × μm));

(iv) $LMIN_\lambda$ – the spectral radiance that is scaled to QCALMIN (W/(m^2 × sr × μm));

(v) for Landsat 7 ETM+ spectral radiance range for band 6 processed after July 1, 2000:

(vi) Low Gain: $LMIN_\lambda$ = 0.0; $LMAX_\lambda$ = 17.04;

(vii) Hight Gain: $LMIN_\lambda$ = 3.2; $LMAX_\lambda$ = 12.65;

(viii) for Landsat 5 TM spectral radiance range for band 6:

(ix) $LMAX_\lambda$ = 15.303; $LMIN_\lambda$ = 1.238

(x) QCAL – the quantized calibrated pixel value in Digital Numbers (DN);

(xi) QCALMAX – the maximum quantized calibrated pixel value (corresponding to

(xii) $LMAX_\lambda$) in DN (DN = 255);

(xiii) QCALMIN – the minimum quantized calibrated pixel value (corresponding to

(xiv) $LMIN_\lambda$) in DN (DN = 1).

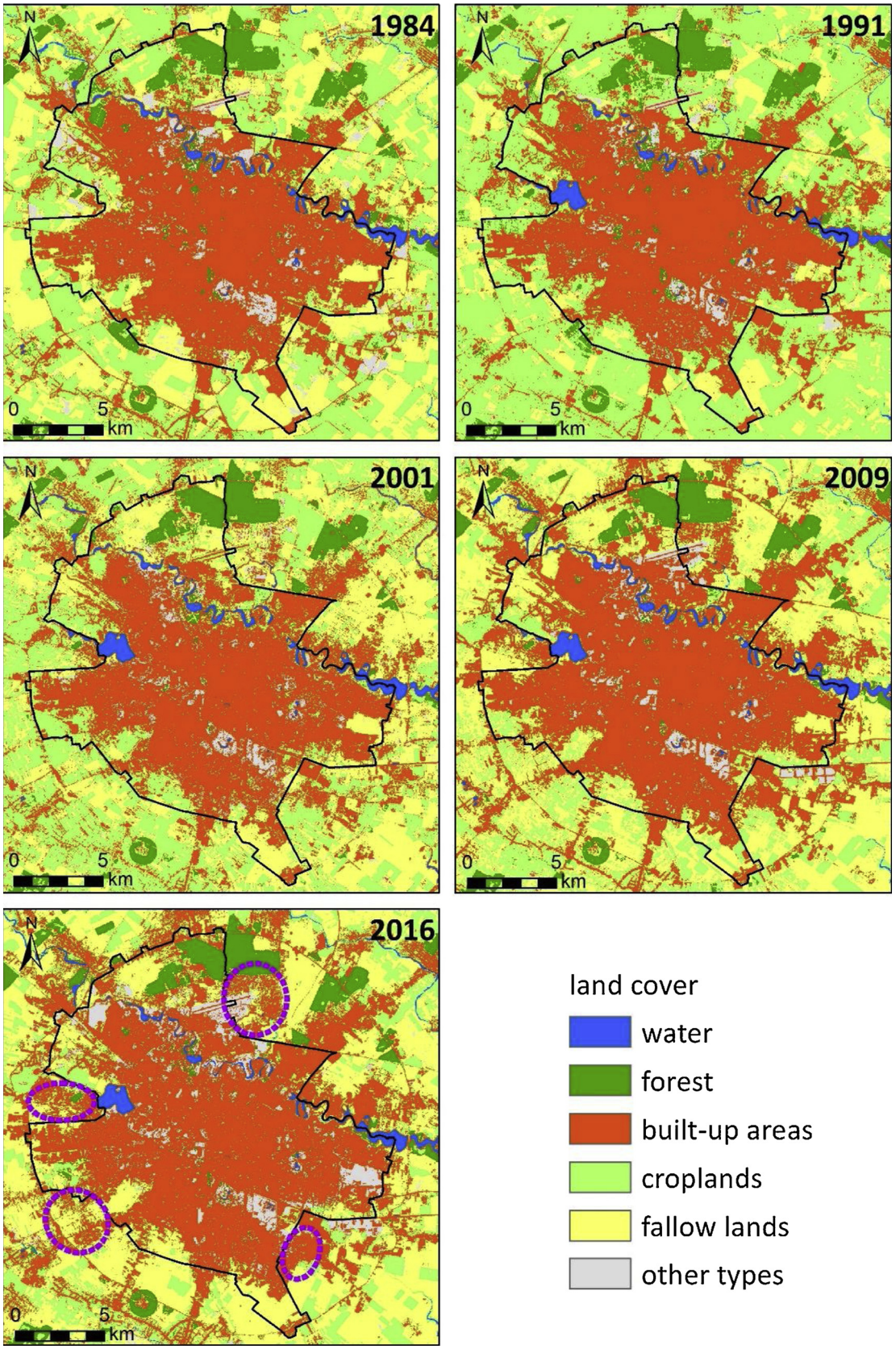


Fig. 2. Land Use/Land cover maps of 1984 (upper left), 1991 (upper right), 2001 (middle left), 2009 (middle right) and 2016 (lower left).

Table 2
Values of the overall accuracy and Kappa coefficient of the classification.

Land cover/accuracy	1984		1991		2001		2009		2016	
	Producer's	User's	Producer's	User's	Producer's	User's	Producer's	User's	Producer's	User's
Water	86	100	90	100	86	97.7	88	97.8	90	100
Forest	100	100	98	98	88	100	92	97.9	92	95.8
Built-up	98	94.2	98	94.2	96	92.3	98	94.2	96	87.3
Croplands	100	90.9	94	88.7	96	73.8	78	79.6	82	87.2
Fallow lands	98	98	86	86	84	93.3	96	84.2	98	89.1
Overall accuracy	96.4		93.2		90		90.4		91.6	
Kappa statistic	95.5		91.6		87.5		88		89.5	

Table 3
Areas covered by each category of land expressed in km^2 and in percentage.

Year/land cover	Water		Forest		Built-up		Croplands		Fallow lands		Other	
	km^2	%	km^2	%	km^2	%	km^2	%	km^2	%	km^2	%
1984	7.7	1.2	35.9	5.7	194.1	31.1	200.3	32.0	165.5	26.5	21.5	3.4
1991	8.5	1.4	36.9	5.9	190.9	30.5	298.8	47.8	75.8	12.1	14.1	2.3
2001	9.8	1.6	28.7	4.6	212.4	34.0	191.4	30.6	166.9	26.7	15.8	2.5
2009	10.7	1.7	30.5	4.9	240.6	38.5	120.1	19.2	209.3	33.5	13.8	2.2
2016	9.3	1.5	31.7	5.1	255.8	40.9	99.4	15.9	201.1	32.2	27.7	4.4
Overall	1.6	0.3	−4.2	−0.7	61.7	9.9	−100.9	−16.1	35.6	5.7	6.2	1.0

(xv) for Landsat 8 (USGS, 2018c), Eq. (3):

$$L_\lambda = M_L \times QCAL + A_L \quad (3)$$

where:

(xvi) L_λ – TOA Spectral Radiance ($W/(m^2 \times sr \times \mu m)$);
(xvii) M_L – Radiance multiplicative scaling factor for the band;
(xviii) A_L – Radiance additive scaling factor for the band;
(xix) QCAL – the quantized calibrated pixel value in Digital Numbers (DN).

In the second step the TOA spectral radiance (L_λ) values is converted into another variable called At-Satellite Brightness Temperature (TB), Eq. (4).

$$TB = K_2/(\ln(K_1/L_\lambda + 1)) \quad (4)$$

where:

TB – At-Satellite Brightness Temperature, in Kelvin (K);
K_1, K_2 – Thermal conversion constants for the band.
Finally, the TOA Brightness Temperature will be convert to land surface temperature values (LST) using the formula (Artis and Carnahan, 1982), Eq. (5):

$$LST = TB/[1 + (\lambda \times TB/\alpha) \times \ln \varepsilon] \quad (5)$$

where:

LST = the Land Surface Temperature in Kelvin (K);
λ – the wavelength of emitted radiance;
$\alpha = hc/k$ (1.438×10^{-2} mK);
h = Planck constant (6.626×10^{-34} J s^{-1}), and c = velocity of light (2.998×108 m s^{-1});
k = Boltzman constant (1.38×10^{-23} J K^{-1});
ε = surface emissivity, calculated according to equation, Eq. (6):

$$\varepsilon = 0.004 \times P_v + 0.986 \quad (6)$$

where P_v is the vegetation proportion calculated following equation, Eq. (7):

$$P_v = [(NDVI - NDVI_{min})/(NDVI - NDVI_{max})]^2 \quad (7)$$

To obtain the land surface temperature values in Celsius (°C), 273.15 was extracted from the initial values (K).

Direct analysis of LST differences for a period of many years would not be appropriate due to seasonal and interannual variability. Therefore, to make the information obtained from satellite images recorded in different years to be comparable, the LST values were normalized, according to the equation, Eq. (8):

$$NLST = (LSTi - LSTmin)/(LSTmax - LSTmin) \quad (8)$$

Where NLST is the normalized LST, LSTi is the initial LST of pixel i, LSTmin and LSTmax are the minimum and maximum LST's value of a given scene.

2.3.4. Retrieval of Normalized Difference Vegetation Index (NDVI)

Because NDVI was used as an indicator for land description (Julien and Sobrino, 2009) and is a less sensitive parameter to changes in atmospheric conditions than other indicators, it has become commonly used in LST studies. In this paper, the NDVI was used to analyze the relationship of LST to the vegetation by linear correlation.

NDVI image was computed using the following formula, Eq. (9):

$$NDVI = \frac{NIR - RED}{NIR + RED} \quad (9)$$

where NIR is near-infrared spectral band and RED is red spectral band, of Landsat.

Values of NDVI are range between −1 and +1, where values close to −1 indicate lack of vegetation, and values close to +1 signify dense vegetation.

2.3.5. Relationship between NDVI and LST

The relationship between LST and NDVI was analyzed through the Pearson correlation coefficient (Eq. (10)), calculated for each land cover class.

The correlation coefficient (r) has values between −1 and +1 with the meaning of: positive/negative association after the coefficient sign and the lack of association for $r = 0$.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{X})(y_i - \bar{Y})}{\sqrt{(\sum_{i=1}^{n}(x_i - \bar{X}))^2 (\sum_{i=1}^{n}(y_i - \bar{Y}))^2}} \quad (10)$$

where $\bar{X}$ and $\bar{Y}$ = mean of variables X and Y; x_i and y_i – variables values with $i = 1, \ldots, n$.

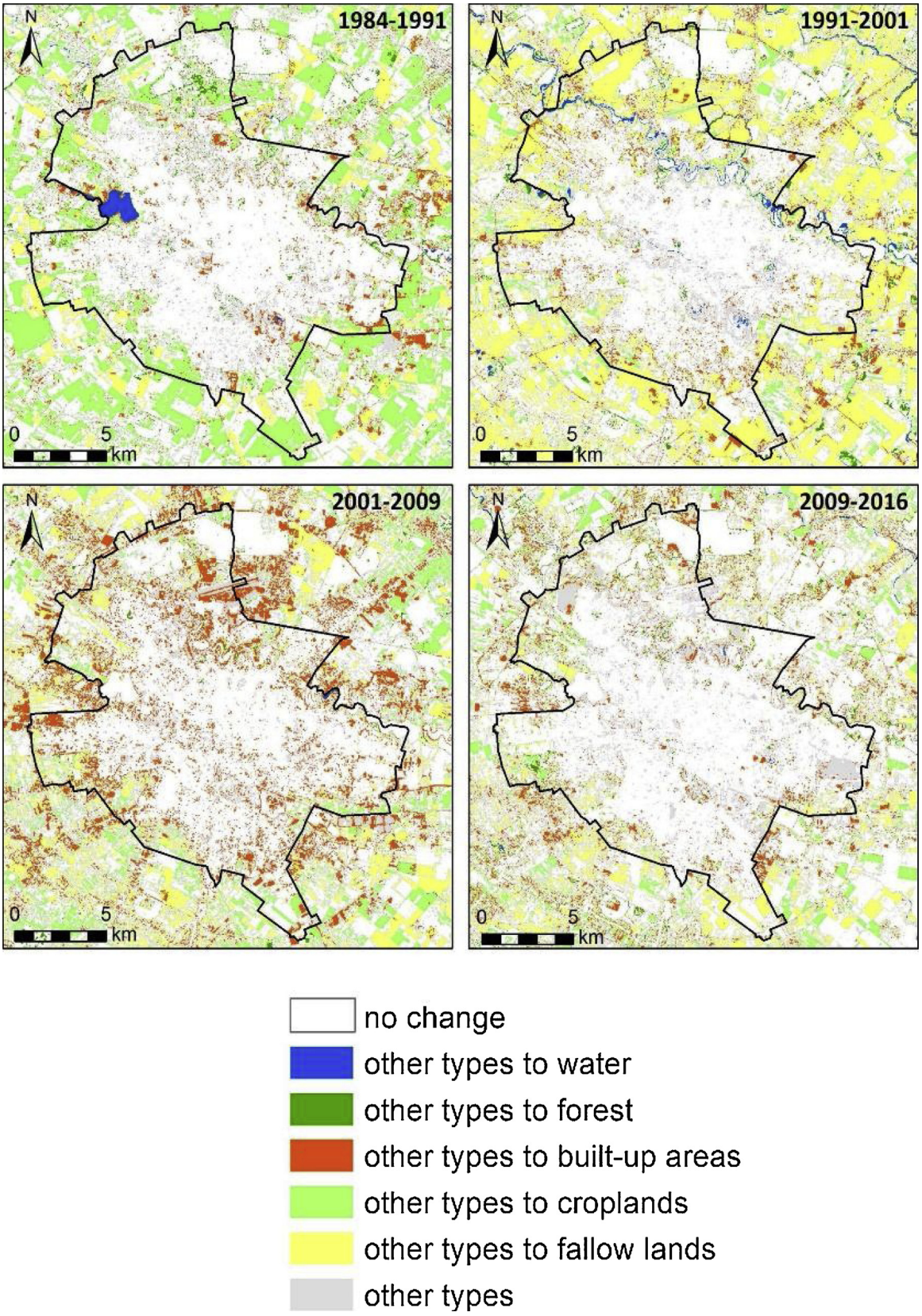


Fig. 3. Land Use/Land Cover changes maps of the study area for 1984 –2016.

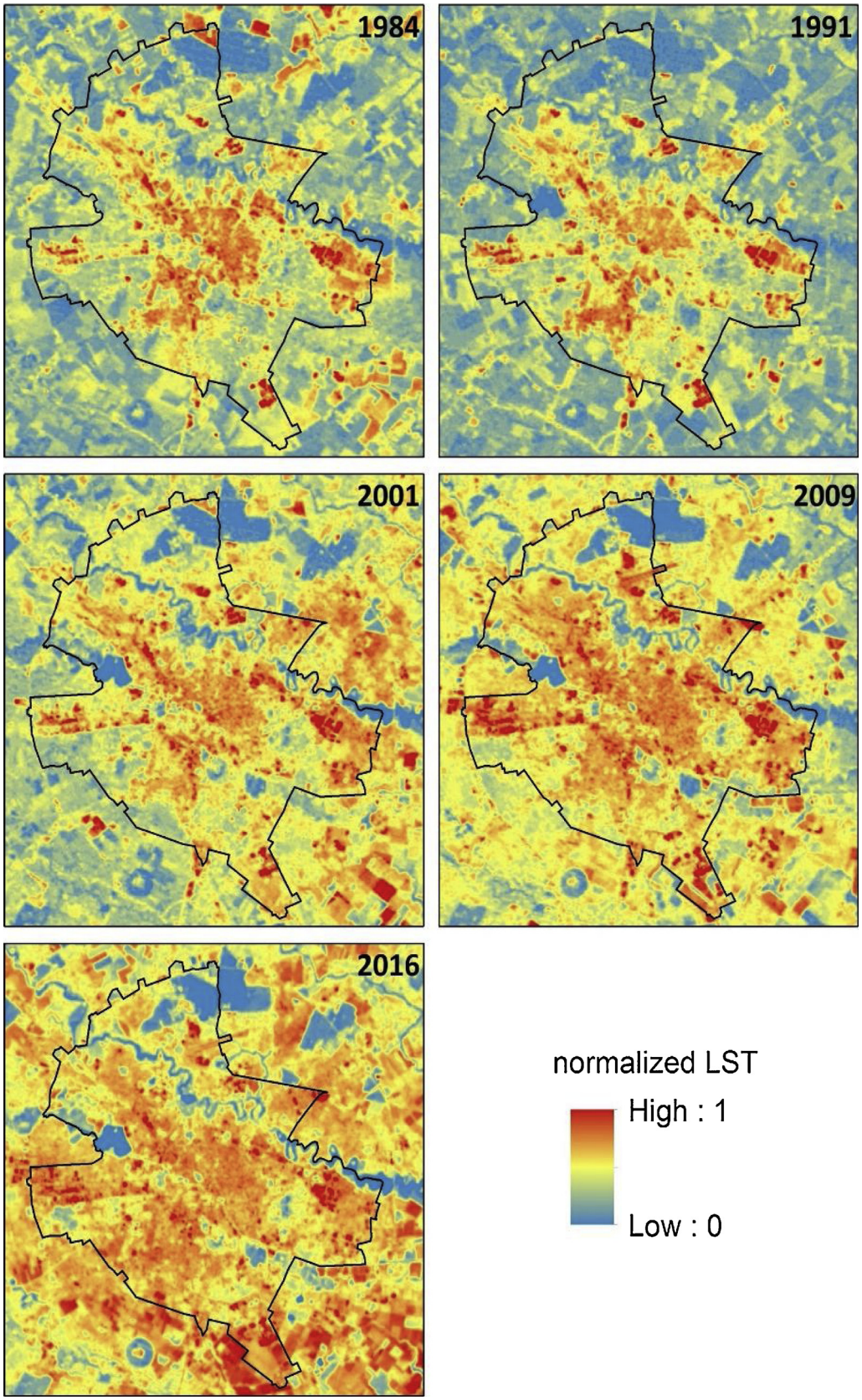


Fig. 4. Normalized land surface temperature maps of Bucharest, 1984 (upper left), 1991 (upper right), 2001 (middle left), 2009 (middle right) and 2016 (lower left).

Table 4
Normalized land surface temperature in different LULC categories, minimum values (MIN), maximum (MAX), average (MEAN) and standard deviation (STDEV).

Year	NST	Water	Forest	Built-up	Croplands	Fallowlands	Other
1984	MIN	0.00	0.10	0.02	0.02	0.07	0.07
	MAX	0.31	0.47	1.00	0.61	0.72	0.83
	MEAN	0.09	0.12	0.34	0.19	0.31	0.30
	STDEV	0.05	0.05	0.11	0.05	0.08	0.09
1991	MIN	0.00	0.03	0.03	0.03	0.08	0.06
	MAX	0.30	0.49	1.00	0.75	0.75	0.54
	MEAN	0.08	0.12	0.34	0.18	0.26	0.26
	STDEV	0.04	0.06	0.12	0.06	0.06	0.07
2001	MIN	0.00	0.05	0.07	0.07	0.12	0.10
	MAX	0.38	0.63	1.00	0.67	0.91	0.70
	MEAN	0.12	0.16	0.43	0.32	0.42	0.36
	STDEV	0.06	0.06	0.10	0.07	0.09	0.08
2009	MIN	0.00	0.16	0.14	0.18	0.10	0.18
	MAX	0.47	0.58	1.00	0.74	0.86	0.70
	MEAN	0.23	0.24	0.50	0.37	0.45	0.44
	STDEV	0.04	0.05	0.09	0.06	0.06	0.10
2016	MIN	0.00	0.14	0.05	0.03	0.07	0.06
	MAX	0.61	0.71	1.00	0.91	0.86	0.91
	MEAN	0.34	0.39	0.60	0.53	0.59	0.57
	STDEV	0.06	0.06	0.07	0.07	0.06	0.08

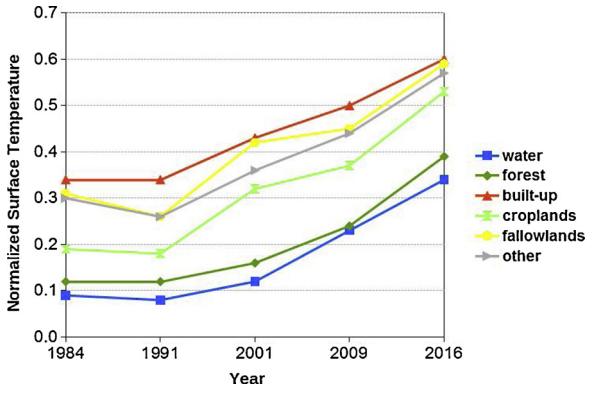


Fig. 5. Relationship between normalized average temperatures of the land surfaces and the six land cover categories and its evolution over time.

3. Results and discussions

3.1. Land cover classification and changes

The results of the supervised classification are presented in Fig. 2. The areas covered by the land cover categories: water, forest, built-up, croplands, fallow lands and other, expressed in km^2 and in percent, for each scene in part, are presented in Table 3.

To verify the accuracy of the image classification, the Accuracy Assessment was performed and the Kappa coefficient was determined.

Values of the Accuracy Assessment (for each land cover class and overall accuracy) and of the Kappa coefficient are shown in Table 2.

Values of the overall accuracy of the land cover classification map are: 96.4% for 1984, 93.2 for 1991, 90% for 2001, 90.4% for 2009 and 91.6 for 2016. The Kappa indices values for 1984, 1991, 2001, 2009 and 2016 maps are: 95.5, 91.6, 87.5, 88, and 89.5. For the all years that are analyzed, these data have reasonably high accuracy (over 90) and are therefore suitable for detecting urban expansion (Lucas et al., 1994).

Table 3 illustrate the evolution of the overall period of 1984–2016 of land cover. After 1984, the area covered by "water" registers an increase due to the appearance of "Lacul Morii" Accumulation Lake, in 1986 (in the NW side of the capital), part of the flood prevention project the canal Morii Lake – Dâmbovița–Văcărești Lake (Fig. 3). The dam was used once and then closed due to high costs of pumping water from Dâmbovița in the lake, so after 1989 the construction works were stopped and the project remained unfinished. Abandoned for several decades, the Concrete Pool of Văcărești Lake was turned by nature into the first urban delta of Europe and thus increasing the "other" covered area as of 2001. The values of the "water" category vary from one scene to another because this class consists largely of accumulation lakes whose level varies according to municipal needs and the amount of precipitations.

The "forest" category remained approximately constant, having below 6% of the analyzed area, in 1984 and 1991, with a slight decrease in the area in 2001. This drop of 1.1% is due to the fact that after 1989 some parts of the forests and lands were restored owners, and later, they have grubbed up to build residential or leisure complexes. Some of the grubbed-up areas have naturally regenerated by 2016, resulting in a 0.3% per cent increase in forest cover between 2001 and 2009, and 0.2% between 2009 and 2016.

The systematization works of Bucharest from the communist-era have involved demolitions that have led to a reduction in the "build-up" area between 1984 and 1991, from 31.1% to 30.5%. After 1990–1991, those greenhouses and other constructions with agricultural or industrial role which remained out of activity after 1989 were demolished. The resulting land was intended for sale for the purpose of construction of new buildings. Thus, the "built-up" area has increased since 1991 from one stage to another by 4% between 1991–2001, 4.5% between 2001–2009 and 2.4% between 2009 and 2016. The decrease in the growth rate of the "built-up" area observed between 2009–2016 is due to the economic crisis that occurred in 2008. The main areas that have undergone changes are those in the north (Otopeni and Băneasa-Pipera), the west (Militari-Chiajna and Bragadiru-Ghencea) and south (Berceni-Popeşti-Leordeni), located near the main access ways of the city (marked with purple in Fig. 2 – lower left). The "croplands" area increased from 32% in 1984 to 47.8% of the area in 1991, since the people re-affiliated with agricultural land after 1989 cultivated these lands. After 2001, the agricultural lands around Bucharest starts to be passed from suburban to intra-urban in order to be purchased by the real estate developers. This leads to a decline in cropland to 30.6% in 2001 and 19.2% in 2009, reaching 15.9% of the area in 2016, which results in the "fallow lands" from 12.1% in 1991 to over 30% by 2016. The "croplands", decreases by 2016 compared to 1984, by 16.1%.

Overlaying of land use/cover maps in pairs (Fig. 3), 1984–1991, 1991–2001, 2001–2009 and 2009–2016, illustrates the spatial dynamics of land use changes presented in Table 3 and discussed previously, highlighting the transformation of cultivated agricultural land into uncultivated agriculture land and later in areas covered by construction.

Overall, between 1984 and 2016, the most significant changes in the areas were those of the built-up categories, which increased by 9.9% and cropland declining by 16.1%. The fallowland category grows by 5.7% until 2016 as compared to 1984. Small changes are recorded by forest and other categories which decrease, respectively, by 0.7% and 1%. Overall, the "water" category increases by 0.3% by 2016, with changes of 0.1% or 0.2% between scenes. The decline of the cropland area (16.1%) is almost entirely found in the growth of the built-up area and the fallowland area (15.6%).

3.2. Effects of land cover change and urbanization on surface temperature

3.2.1. Relationship between land cover types and surface temperature

In order to determine the relationship between land cover changes and surface temperature, first it was necessary to study the thermal signature of each land category and then to overlay the land cover changes maps with those of surface temperatures.

Since after the conversion of satellite data into LST, the temperature scale of each scene was different; to visualize the impact of land use

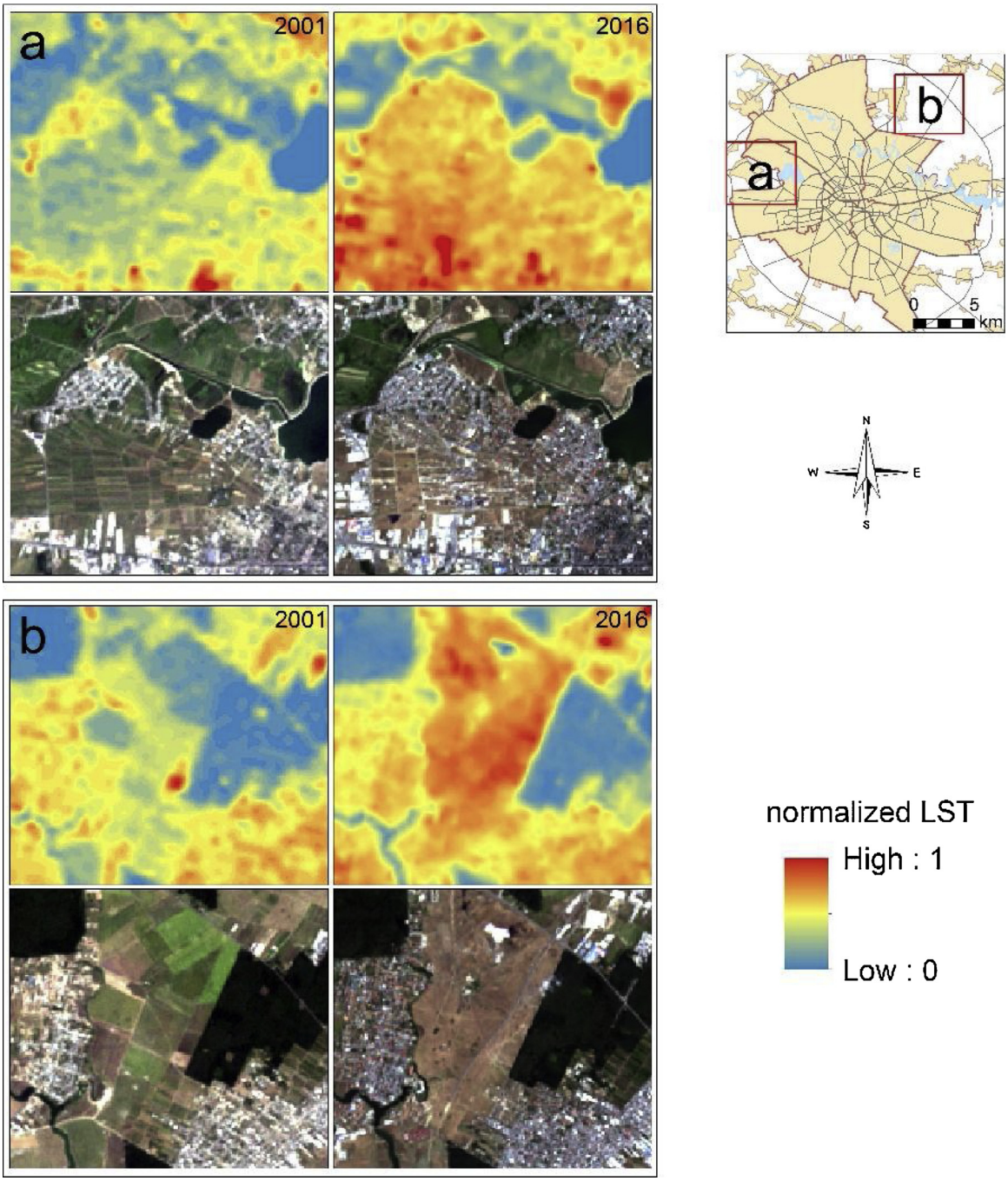


Fig. 6. Examples of changes in surface coverage and their influence on surface temperatures for Militari-Chiajna (up) and Pipera-Voluntari (bottom), August 4, 2001 (left), and August 5, 2016 (right).

changes on surface temperature, the surface temperatures values were normalized to range from 0 to 1. The spatial distribution of normalized surface temperatures is shown in Fig. 4 and highlights the areas with higher temperatures values (over 0.5) representing the normalized surface temperature, marked on the map with yellow, therefore expansion of the Bucharest's Urban Heat Island.

For each land cover category, there were determined normalized land surface temperatures (NLST): minimum (MIN), maximum (MAX), average (MEAN), and standard deviation (STDEV). These values, corresponding to each analyzed scenario and are presented in Table 4. The lowest standard deviation value corresponds to water bodies (0.04), as they have a relatively narrow range of temperature variation, with lower deviations from the average. The highest standard deviation value is obtained for the built-up category (0.12), and can be caused by urban structure, and also because the building materials have different degrees of heating and reflectivity, so the deviation from the average temperature is higher. The smallest values (on average), referring to the 1984 scene, correspond to the water bodies (0.09), forest (0.12) and the croplands (0.19), while the highest values correspond to the fallowlands (0.31) and built-up (0.34). Also, high values correspond to the other category (0.3) that is characterized by areas under construction (construction sites – so wasteland mixed with built-up areas), landfill, wetlands.

The relationship between the normalized surfaces temperatures and the six land cover categories is illustrated in Fig. 5. Following the evolution over time (Fig. 5) it is observed that the order of these

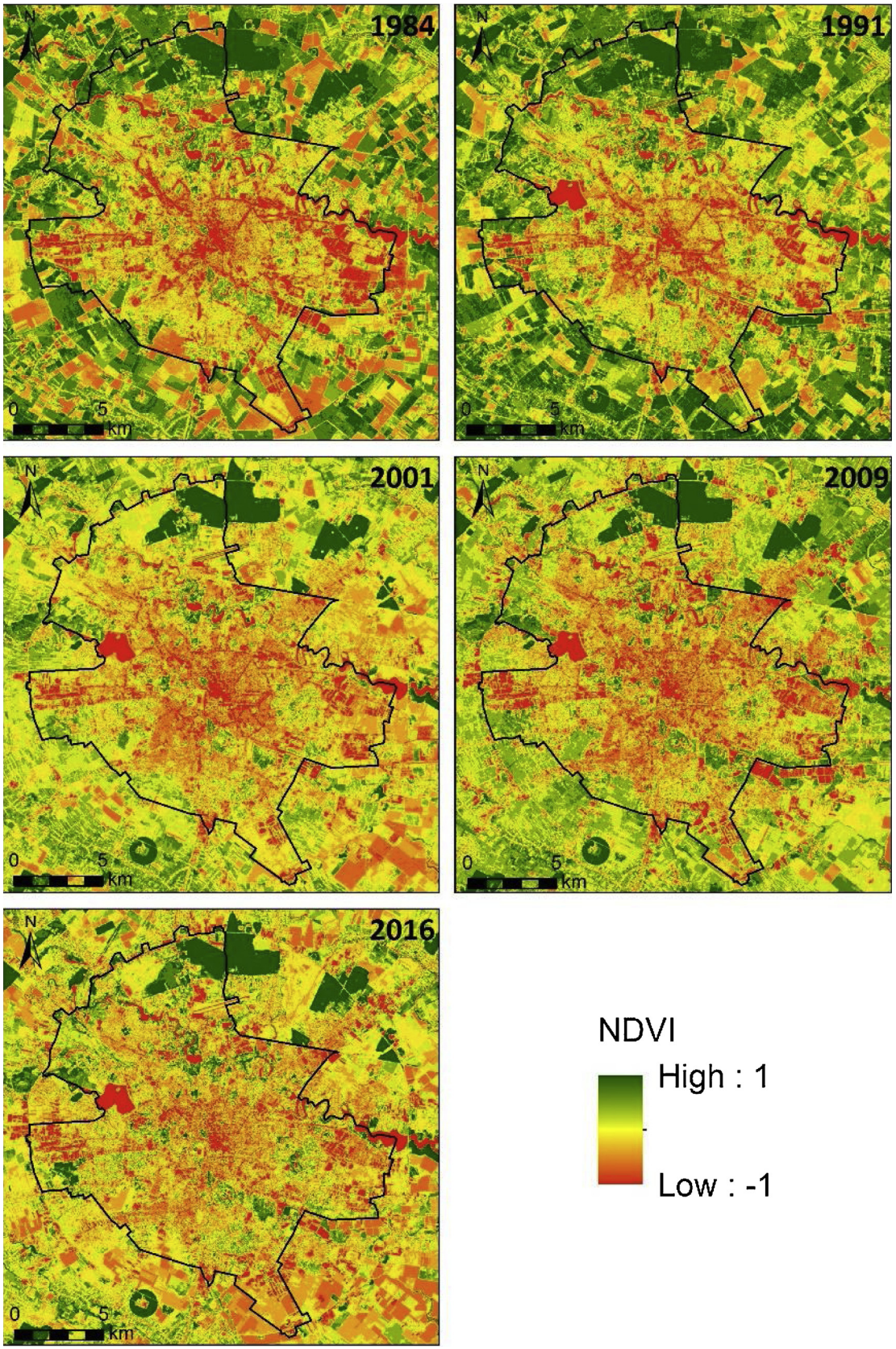


Fig. 7. NDVI classes of 1984 (upper left), 1991 (upper right), 2001 (middle left), 2009 (middle right) and 2016 (lower left).

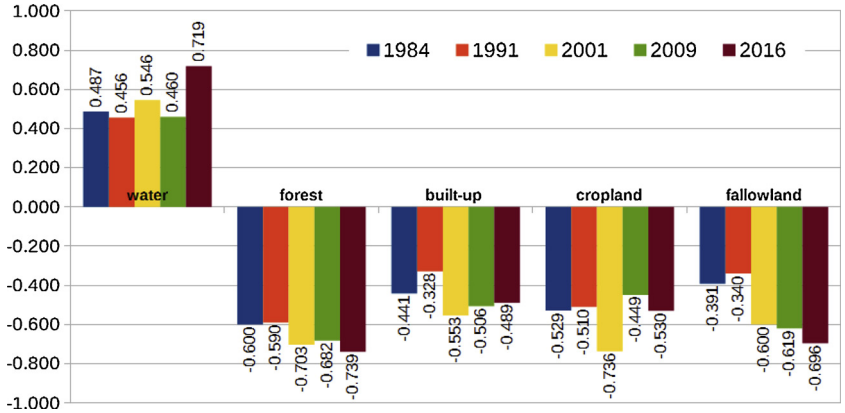


Fig. 8. Relationship between land surface temperature (LST) and vegetation index (NDVI), investigated in the performance of the Pearson's correlation coefficients.

thermal signatures of the surfaces is maintained, but the values increase relatively constant from one scene to another, with small exceptions being found in the cropland and fallow land categories depending on the type of the crops and the plant vegetative stage. The upward evolution in time of normalized surface temperatures can be explained as an effect of expanding and enhancing the urban heat island due to the land cover changes and the growth of built-up areas in the general context of climate change. The phenomenon has been emphasized since 1991 (the inflection points on the chart (in Fig. 5) when the agricultural lands around Bucharest are no longer being cultivated and their role is changing to land for the construction of buildings, implicitly for construction sites.

From the land use/land cover maps overlapping with land surface temperature maps it is noticeable that the surfaces that have changed their use and becoming built-up and fallowlands, are the ones whose temperature has grown the most. Detailed examples of these types of land use changes and their surface temperature effects for the Militari-Chiajna and Pipera-Voluntari areas are illustrated in Fig. 6. From these examples it can be seen that the temperature on the surfaces that have changed their use from cultivated land to waste land or uncovered construction sites, the temperature can increase by up to 0.2 (in normalized values), while those that have changed to areas with buildings the temperature can increase by up to 0.5.

As a consequence of these changes in LULC and LST, it was found that both qualitative (visualizing the maps in Fig. 4) and quantitative (calculating the difference between the average of surface temperature within the boundary limits between the administrative area of Bucharest Municipality and the average of surface temperature from the surrounding areas), strictly for the analyzed area, the UHI intensity drops from 0.08 in 1984 to 0.01 in 2016 (NST). In the broader context, these results mean the decrease of the thermal difference between urban and peri-urban areas and the extension of the area where the Bucharest's UHI manifests.

3.2.2. Relationship between NDVI and LST

The relationship between the surface temperature (Fig. 4) and the NDVI (Fig. 7) was analyzed on the pair data sets (LST and NDVI values corresponding to the same pixel) and included a number of 738,077 points. For each year, there were calculated correlation coefficients for each land use category. The values of the Pearson correlation coefficient (Fig. 8) indicate a negative correlation of land surface temperature with NDVI for all land cover types for all years. For water-covered areas, positive correlation coefficients (between 0.456 in 1991 and 0.719 in 2016) were obtained because the water has low values for both NDVI (negative values) and for LST. The strongest negative correlations were obtained for the forest category (0.739 in 2016 and 0.703 in 2001) and the cropland category (0.736 in 2001). The lowest negative correlations were obtained for the built-up category (0.328 in 1991) and fallowland (0.340 in 1991 and 0.391 in 1984).

The relationship between LST and NDVI suggests that the higher is the density of the vegetation that covers the surface, the lower is the temperature for the surface area, because higher NDVI causes high evapotranspiration.

4. Conclusions

In this study, an analysis based on multi-temporal remote sensing data was carried out to study the land use changes and their influence on land surface temperature over the period 1984–2016. There was realized a supervised classification of the analyzed area in 6 land use/cover classes: water, forest; built-up, cropland, fallow land and other with high accuracy (over 90% overall accuracy).

Throughout the analyzed period, there have been found significant land use changes due to the political and economic context of the period, as well as a dynamic and interdependence for the cropland, fallow land and built-up land categories were found. Overall, the result of these changes indicates the decrease for the cropland by 16.1% and for the forest area by 0.7% and the increase of the built-up area by 9.9%, the fallow land area by 5.7%, the other area by 1% and the water area by 0.3%. It can be notice a decrease in the cultivated land areas that first turn into fallow lands to eventually become built-up areas. The most dramatic changes have been found in the N and W areas of the city, but also all the areas located near the main traffic roads.

The study also highlighted that land use changes and land cover due to anthropogenic activities lead to changes in LST. The determination of the thermal signatures of the different categories of land in the area underlined the contribution of each of them to maintaining and expanding the phenomenon of “Urban Heat Island”. It was obtained an increase in LST from one scene to another (for all classes of terrain) and a decrease in the difference between the average of land surface temperature inside and outside the city, from 0.08 in 1984 to 0.01 in 2016 (expressed in surface temperatures normalized values), suggesting the expansion of the area affected by Bucharest's UHI.

From the analysis of the vegetation index, the tendency of diminishing the vegetation areas was observed. A negative relationship between NDVI and LST was obtained for vegetation-covered areas and a positive one for water bodies, as LST is strongly influenced by LCLU and vegetation, low LST values being obtained for dense vegetation areas, and for those covered by water bodies.

The results presented in the study suggest the necessity of an appropriate land use planning to mitigate the temperatures increases in urban area.

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